

PSLE MATHEMATICS PRACTICE PAPER 3

Area, Perimeter & Circumference

Duration: 1 hour

Total Marks: 50

Name: _____

Date: _____

SECTION A: Short Answer Questions (20 marks)

1. A square has side 7 cm. Find its area. (2 marks)

2. A rectangle has length 12 cm and width 5 cm. Find its perimeter. (2 marks)

3. A triangle has base 8 cm and height 6 cm. Find its area. (2 marks)

4. A circle has radius 5 cm. Find its circumference. (Use $\pi = 3.14$) (2 marks)

5. The area of a square is 64 cm². Find its side length. (2 marks)

6. A rectangle has perimeter 30 cm and length 10 cm. Find its width. (2 marks)

7. Find the area of a parallelogram with base 8 cm and height 4 cm. (2 marks)

8. A circle has diameter 14 cm. Find its area. (Use $\pi = 22/7$) (4 marks)

SECTION B: Long Answer Questions (30 marks)

9. A composite shape consists of a rectangle ($10\text{ cm} \times 6\text{ cm}$) and a triangle (base 10 cm , height 4 cm) on top. Find the total area. (4 marks)

10. A square garden has side 20 m . It needs to be fenced. If fencing costs $\$5$ per meter, what is the total cost? (3 marks)

11. An L-shaped figure is made of two rectangles. Rectangle 1: $8\text{ cm} \times 3\text{ cm}$, Rectangle 2: $4\text{ cm} \times 5\text{ cm}$. Find the total perimeter. (4 marks)

12. The circumference of a circle is 44 cm . Find its radius. (Use $\pi = 22/7$) (3 marks)

13. A trapezium has parallel sides of 6 cm and 10 cm, and height 5 cm. Find its area. (3 marks)

14. A rectangular field is 50 m long and 30 m wide. Find both its perimeter and area. (4 marks)

15. Two circles have radii 3 cm and 4 cm. Find the difference in their areas. (Use $\pi = 3.14$) (4 marks)

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SOLUTIONS

SECTION A SOLUTIONS

Q1: Square area

$$\text{Area} = \text{side}^2 = 7^2 = 49 \text{ cm}^2$$

Q2: Rectangle perimeter

$$P = 2(l + w) = 2(12 + 5) = 34 \text{ cm}$$

Q3: Triangle area

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 6 = 24 \text{ cm}^2$$

Q4: Circle circumference

$$C = 2\pi r = 2 \times 3.14 \times 5 = 31.4 \text{ cm}$$

Q5: Square side from area

$$\text{Side} = \sqrt{64} = 8 \text{ cm}$$

Q6: Rectangle width

$$P = 2(l + w), 30 = 2(10 + w), 15 = 10 + w, w = 5 \text{ cm}$$

Q7: Parallelogram area

$$\text{Area} = \text{base} \times \text{height} = 8 \times 4 = 32 \text{ cm}^2$$

Q8: Circle area

$$A = \pi r^2 = \frac{22}{7} \times 7^2 = 22 \times 7 = 154 \text{ cm}^2$$

SECTION B SOLUTIONS

Q9: Composite shape

$$\text{Rectangle area} = 10 \times 6 = 60 \text{ cm}^2, \text{ Triangle area} = \frac{1}{2} \times 10 \times 4 = 20 \text{ cm}^2, \text{ Total} = 80 \text{ cm}^2$$

Q10: Fencing cost

$$\text{Perimeter} = 4 \times 20 = 80 \text{ m}, \text{ Cost} = 80 \times \$5 = \$400$$

Q11: L-shaped perimeter

$$\text{Outer perimeter} = 2(8+3) + 2(4) + 2(5-3) + 2(8-4) = 38 \text{ cm}$$

Q12: Circle radius from circumference

$$C = 2\pi r, 44 = 2 \times \frac{22}{7} \times r, r = 7 \text{ cm}$$

Q13: Trapezium area

$$\text{Area} = \frac{1}{2} \times (a + b) \times h = \frac{1}{2} \times (6 + 10) \times 5 = 40 \text{ cm}^2$$

Q14: Field perimeter and area

$$\text{Perimeter} = 2(50 + 30) = 160 \text{ m}, \text{Area} = 50 \times 30 = 1500 \text{ m}^2$$

Q15: Difference in circle areas

$$\text{Area}_{\blacksquare} = 3.14 \times 3^2 = 28.26 \text{ cm}^2, \text{Area}_{\blacksquare} = 3.14 \times 4^2 = 50.24 \text{ cm}^2, \text{Difference} = 21.98 \text{ cm}^2$$

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